### 6.4.6 Difference between prokaryotic and eukaryotic translation

| S. No | Prokaryotic translation | Eukaryotic translation |
|---|---|---|
| 1 | Transcription and translation are part of a continuous process and occurs simultaneously | Transcription and translation are separate process |
| 2 | The 5' end of the transcribed mRNA is immediately available for translation without any modification | The primary transcript is subjected to post transcriptional modifications and has to cross the nuclear membrane to reach the cytoplasam for translation |
| 3 | 70S ribosome is involved, with a 50S larger subunit and a 30S smaller sub-unit | 80S ribosome is involved with a 60S larger subunit and a 40S smaller subunit |
| 4 | Larger subunit of ribosome is made of 23S, 5S rRNA and 36 different proteins | Larger subunit is made of 28S, 5S, 5.8S rRNA and 40 different proteins |
| 5 | Smaller subunit of ribosome is composed of 16S rRNA and 21 different proteins | Smaller subunit of ribosome is made of 18S rRNA and 33 proteins |
| 6 | mRNA is polycistronic | mRNA is monocystronic |

## 6.5 RECOMBINANT DNA TECHNOLOGY

This technology is also known as DNA cloning allows us to insert a DNA fragment of interest such as a new gene into an organism, which doesn't contain that fragment of DNA. Though there are a variety of strategies to achieve this, the simplest of it is introducing a new gene into bacteria through plasmid vectors. Plasmids are small circular DNA molecules present in bacteria. They can be used to carry a foreign gene into the bacteria and hence the name vectors. The entire process is explained in figure 18. and consists of the following important steps

1. A bacterial plasmid designed with an antibiotic resistant marker gene is selected as a vector.
2. The circular plasmid vector is cleaved at specific sites using special enzymes called restriction endonuclease which are enzymes that cleave DNA at specific sequence patterns. For example EcoRI is a restriction enzyme which cleaves at the sequence GAATC.
3. The gene to be cloned is inserted in the gap produced by the endonuclease and an enzyme called DNA ligase is used to join the DNA pieces. This results in a circular plasmid vector containing the gene of interest or the recombinant plasmid.
4. This recombinant plasmid vector is now transferred into the cell by means of a process called transformation.



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